Ethan N. W. Epperly

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Education

California Institute of Technology *Pasadena, CA*

PhD Applied and Computational Mathematics *September 2020 – June 2025*

- Thesis: *Make the most of what you have: Resource-efficient randomized algorithms for matrix computations*
- Advisor: Prof. Joel A. Tropp

University of California, Santa Barbara *Santa Barbara, CA*

BS Mathematics and BS Computing *September 2016 – June 2020*

Professional Experience

2025–2028	**Miller Research Fellow**, University of California, Berkeley
2023–2025	**PhD Candidate**, California Institute of Technology
2020–2023	**PhD Student**, California Institute of Technology
2021	**Graduate Student Research Intern**, Lawrence Berkeley National Laboratory
2020	**Graduate Student Research Intern**, Lawrence Livermore National Laboratory
2019	**Undergraduate Research Assistant**, University of California, Santa Barbara
2015–2019	**Research Intern**, Sandia National Laboratories

Publication List

Published or accepted

T. Chen, **E. N. Epperly**, R. A. Meyer, C. Musco, & A. Rao (2026). Does block size matter in randomized block Krylov low-rank approximation?. Proceedings of the 2026 Annual ACM-SIAM Symposium on Discrete Algorithms, 1026–1046. (preprint)

C. Camaño, **E. N. Epperly**, & J. A. Tropp (2026). Successive Randomized Compression: A Randomized Algorithm for the Compressed MPO-MPS Product. Quantum 10, 2022. (preprint)

M. Dereziński, **E. N. Epperly**, & R. A. Meyer (2026). The Matrix-Vector Complexity of $Ax = b$. Proceedings of the 2026 Conference on Learning Theory, accepted. (preprint)

E. N. Epperly, A. Greenbaum, & Y. Nakatsukasa (2026). Stable Algorithms for General Linear Systems by Preconditioning the Normal Equations. Numerische Mathematik. (preprint)

E. N. Epperly, G. Goldshlager, & R. J. Webber (2026). Randomized Kaczmarz with Tail Averaging. Applied and Computational Harmonic Analysis 80, 101812. (preprint)

E. N. Epperly, M. Meier, & Y. Nakatsukasa (2026). Fast Randomized Least-Squares Solvers Can Be Just as Accurate and Stable as Classical Direct Solvers. Communications on Pure and Applied Mathematics 79, 293–339. (preprint)

E. N. Epperly (2026). Adaptive Randomized Pivoting and Volume Sampling. SIAM Journal on Matrix Analysis and Applications, accepted. (preprint)

Y. Chen, **E. N. Epperly**, J. A. Tropp, & R. J. Webber (2025). Randomly Pivoted Cholesky: Practical Approximation of a Kernel Matrix with Few Entry Evaluations. Communications on Pure and Applied Mathematics 78, 995–1041. (preprint)

E. N. Epperly, J. A. Tropp, & R. J. Webber (2025). Embrace Rejection: Kernel Matrix Approximation by Accelerated Randomly Pivoted Cholesky. SIAM Journal on Matrix Analysis and Applications, 2527–2557. (preprint)

H. Wilber, **E. N. Epperly**, & A. H. Barnett (2025). Superfast Direct Inversion of the Nonuniform Discrete Fourier Transform via Hierarchically Semiseparable Least Squares. SIAM Journal on Scientific Computing, A1702-A1732. (preprint)

Z. Ding, **E. N. Epperly**, L. Lin, & R. Zhang (2024). The ESPRIT Algorithm under High Noise: Optimal Error Scaling and Noisy Super-Resolution. 2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS), 2344–2366. (preprint)

E. N. Epperly (2024). Fast and Forward Stable Randomized Algorithms for Linear Least-Squares Problems. SIAM Journal on Matrix Analysis and Applications, 1782–1804. (preprint)

E. N. Epperly & J. A. Tropp (2024). Efficient Error and Variance Estimation for Randomized Matrix Computations. SIAM Journal on Scientific Computing 46, A508-A528. (preprint)

E. N. Epperly, J. A. Tropp, & R. J. Webber (2024). XTrace: Making the Most of Every Sample in Stochastic Trace Estimation. SIAM Journal on Matrix Analysis and Applications, 1–23. (preprint)

E. Epperly & E. Moreno (2023). Kernel Quadrature with Randomly Pivoted Cholesky. Advances in Neural Information Processing Systems 36, 65850–65868. (preprint)

E. N. Epperly, L. Lin, & Y. Nakatsukasa (2022). A Theory of Quantum Subspace Diagonalization. SIAM Journal on Matrix Analysis and Applications 43, 1263–1290. (preprint)

N. Govindarajan, **E. N. Epperly**, & L. D. Lathauwer (2022). $(L_r, L_r, 1)$-decompositions, Sparse Component Analysis, and the Blind Separation of Sums of Exponentials. SIAM Journal on Matrix Analysis and Applications 43, 912–938.

E. N. Epperly, N. Govindarajan, & S. Chandrasekaran (2021). Minimal Rank Completions for Overlapping Blocks. Linear Algebra and its Applications 627, 185–198. (preprint)

E. N. Epperly & R. B. Sills (2020). Transient Solute Drag and Strain Aging of Dislocations. Acta Materialia 193, 182–190.

E. N. Epperly & R. B. Sills (2020). Comparison of Continuum and Cross-Core Theories of Dynamic Strain Aging. Journal of the Mechanics and Physics of Solids 141, 103944.

Under review

E. N. Epperly, T. Park, & Y. Nakatsukasa (2026). Fast, High-Accuracy, Randomized Nullspace Computations for Tall Matrices. arXiv preprint arXiv:2602.16797v1.

E. N. Epperly & R. J. Webber (2026). Sharp Analysis of Sketched Least Squares and Randomized Low-Rank Approximation. arXiv preprint arXiv:2605.19096v1.

C. Camaño, **E. N. Epperly**, R. A. Meyer, & J. A. Tropp (2026). Linear Algebra at Exponential Scale via Tensor Network Dimension Reduction. arXiv preprint arXiv:2606.15350v1.

C. Camaño, **E. N. Epperly**, R. A. Meyer, & J. A. Tropp (2025). Faster Linear Algebra Algorithms with Structured Random Matrices. arXiv preprint arXiv:2508.21189v1.

M. Díaz, **E. N. Epperly**, Z. Frangella, J. A. Tropp, & R. J. Webber (2023). Robust, Randomized Preconditioning for Kernel Ridge Regression. arXiv preprint arXiv:2304.12465v5.

Reports and other

N. Amsel, Y. Baumann, P. Beckman, P. Bürgisser, C. Camaño, T. Chen, E. Chow, A. Damle, M. Derezinski, M. Embree, **E. N. Epperly**, R. Falgout, M. Fornace, A. Greenbaum, C. Greif, D. Halikias, Z. Huang, E. Jarlebring, Y. Koutis, D. Kressner, R. Kyng, J. Liesen, J. Lok, R. A. Meyer, Y. Nakatsukasa, K. Pearce, R. Peng, D. Persson, E. Rebrova, R. Schneider, R. Shah, E. Solomonik, N. Srivastava, A. Townsend, R. J. Webber, & J. Williams (2026). Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop. arXiv preprint arXiv:2602.05394v2.

E. N. Epperly (2025). Make the Most of What You Have: Resource-efficient Randomized Algorithms for Matrix Computations. PhD dissertation, California Institute of Technology.

E. N. Epperly, A. T. Barker, & R. D. Falgout (2020). Smoothers for Matrix-Free Algebraic Multigrid Preconditioning of High-Order Finite Elements. Technical report LLNL-TR-814531, Lawrence Livermore National Lab. (LLNL), Livermore, CA (United States).

S. Chandrasekaran, **E. Epperly**, & N. Govindarajan (2019). Graph-Induced Rank Structures and Their Representations. arXiv preprint arXiv:1911.05858v3.

D. K. Ward, X. Zhou, R. A. Karnesky, R. Kolasinski, M. E. Foster, K. Thurmer, P. Chao, **E. N. Epperly**, J. A. Zimmerman, B. M. Wong, & R. B. Sills (2015). Understanding H Isotope Adsorption and Absorption of Al-alloys Using Modeling and Experiments (LDRD: #165724). Technical report SAND-2015-8388, Sandia National Lab. (SNL-CA), Livermore, CA (United States).

Awards, Fellowships, & Grants

2026 **SIGEST Award**, SIAM *(for "A Theory of Quantum Subspace Diagonalization")*
2025 **SIAM Student Paper Prize**, SIAM *(for "A Theory of Quantum Subspace Diagonalization")*
2025 **Carey & Co. Prize in Applied Mathematics**, California Institute of Technology
2025 **Mathematical Sciences Postdoctoral Research Fellowship**, NSF *(declined in favor of Miller)*
2020–2024 **Computational Science Graduate Fellowship**, Department of Energy
2023 **ICIAM 2023 Student Travel Award**, SIAM
2022 **Thomas A. Tisch Prize for Graduate Teaching in CMS**, Caltech CMS Department
2020 **Raymond L Wilder Award**, UCSB Math Department
2020 **Chancellor's Award for Excellence in Undergraduate Research**, UCSB
2020 **Graduate Research Fellowship**, NSF *(declined in favor of CSGF)*
2020 **Hertz Foundation Fellowship Finalist**, Hertz Foundation
2016 **Achievement in Mathematics Award**, Las Positas College

Presentations

Talks

November 2026. *Could a randomized algorithm be better than Krylov iteration?* SIAM Mathematics of Data Science Conference, Salt Lake City, Utah, USA.

May 2026. *Computational linear algebra from time evolution and noisy inner products*. Conference of the International Linear Algebra Society, Blacksburg, Virginia, USA.

April 2026. *Sparser, better, faster, stronger: Emerging approaches in the theory and practice of randomized dimensionality reduction*. Applied Math Seminar, Seattle, Washington, USA.

February 2026. *Sparser, better, faster, stronger: Emerging approaches in the theory and practice of randomized dimensionality reduction*. Princeton IDeAS Seminar, Princeton, New Jersey, USA.

February 2026. *When is a random linear map injective?* UC Davis Probability Seminar, Davis, California, USA.

February 2026. *Sparser, better, faster, stronger: Emerging approaches in the theory and practice of randomized dimensionality reduction*. UC Berkeley / Lawrence Berkeley Lab Applied Math Seminar, Berkeley, California, USA.

February 2026. *Super-resolution and quantum eigensolvers*. UC Davis Math of Data and Decisions (MADD) Seminar, Davis, California, USA.

February 2026. *What is the role of Monte Carlo in randomized linear algebra?* Monte Carlo Online Seminar.

October 2025. *Column subset selection, active learning, and determinantal point processes.* 1W-MINDS Seminar Series.

October 2025. *Does block size matter in randomized block Krylov iteration?* Simons Institute Workshop on Linear Systems and Eigenvalue Problems, Berkeley, California, USA.

September 2025. *Why should I care about sketching?* Simons Institute "Why Should I Care About X?" Seminar Series, Berkeley, California, USA.

July 2025. *Krylov methods on quantum computers: Challenges and opportunities*. ***Invited SIAM student paper prize talk.*** SIAM Annual Meeting, Montreál, Quebec, Canada.

July 2025. *Preconditioning for linear regression and kernel methods: The state of play*. International Conference on Continuous Optimization, Los Angeles, California, USA.

June 2025. *Randomly pivoted Cholesky: Near-optimal positive semidefinite low-rank approximation from a small number of entry evaluations*. Householder Symposium, Ithaca, New York, USA.

May 2025. *Successive randomized compression: A randomized algorithm for the compressed MPO–MPS product*. Tensor Network Weekly Seminar Series, Google.

April 2025. *Randomly pivoted Cholesky: Fast, accurate matrix approximation for scientific machine learning*. Claremont Center for Mathematical Sciences Applied Math Seminar, Pomona, California, USA.

February 2025. *Super-resolution and quantum eigensolvers*. UC San Diego Seminar on the Mathematics of Information, Data, and Signals, San Diego, California, USA.

January 2025. *Could a randomized least-squares solver be the default in your favorite programming language?* Joint Mathematics Meeting, Seattle, Washington, USA.

October 2024. *Sparse random embeddings for reliable matrix computations*. AMS Fall Western Sectional Meeting, Riverside, California, USA.

October 2024. *Embrace rejection: Faster low-rank approximation by rejection sampling-accelerated randomly pivoted Cholesky*. SIAM Mathematics of Data Science Conference, Atlanta, Georgia, USA.

July 2024. *Randomly pivoted Cholesky: Faster matrix approximation for scientific machine learning*. Department of Energy Computational Science Graduate Fellowship Program Review, Washington DC, USA.

April 2024. *Could a randomized least-squares solver become the default in your favorite programming language?* NYU Algorithms & Theory Group, New York, New York, USA.

April 2024. *Could a randomized least-squares solver become the default in your favorite programming language?* Stanford Applied Math Seminar, Palo Alto, California, USA.

December 2023. *Does sketching work?* UC Irvine Applied & Computational Math Seminar, Irvine, California, USA.

September 2023. *Randomly pivoted Cholesky: Simple, effective positive semidefinite low-rank approximation*. UC Berkeley / Lawrence Berkeley Lab Applied Math Seminar, Berkeley, California, USA.

August 2023. *How and why to "uncompute" the randomized SVD with applications to trace and variance estimation*. International Congress of Industrial and Applied Mathematics, Tokyo, Japan.

July 2023. *Quantum Krylov methods: Why do they work?*. IBM Advanced Discovery Seminar, IBM.

April 2023. *XTrace: Estimating the Trace of a Large Matrix Without Reading Its Entries*. Invited presentation at the San Francisco State University SIAM Chapter, San Fransisco, California, USA.

March 2023. *Quantum Krylov methods: Why do they work?* SIAM Computational Science and Engineering Conference, Amsterdam, The Netherlands.

September 2022. *Matrix jackknife variability estimation*. SIAM Mathematics of Data Science Conference, San Diego, California, USA.

May 2022. *Quantum subspace diagonalization: A numerical analysis perspective*. Presentation. Southern California Applied Mathematics Symposium, Claremont, Calfornia, USA.

March 2019. *Towards a fully algebraic generalization of SSS to 2D operators*. SIAM Computational Science and Engineering Conference, Spokane, Washington, USA.

Posters

December 2023. *Kernel quadrature with randomly pivoted Cholesky*. ***NeurIPS spotlight poster.*** New Orleans, Louisiana, USA.

March 2022. *A theory of quantum subspace diagonalization*. Quantum Information Processing, Pasadena, Calfornia, USA.

Teaching Experience

Spr. 2025	**Teaching assistant**, ACM 207: Matrix analysis
Win. 2024	**Teaching assistant**, ACM 217: Advanced topics in probability: Random matrix theory
Sept. 2023	**Mathematics refresher**, Co-taught week-long course for incoming PhD students
Sept. 2022	**Mathematics refresher**, Co-taught week-long course for incoming PhD students
Sept. 2021	**Mathematics refresher**, Designed, co-taught week-long course for incoming PhD students

Mentoring

Sum. 2023	**Christian Camaño**, primary mentor. Mentored this SF state undergraduate on a research project concerning randomized algorithms for tensor network contractions

Outreach & Professional Development

Organizing

2026	**Conference of the International Linear Algebra Society, *Polynomials, Krylov Methods and Applications***, Co-organizer
2025–2026	**UC Berkeley / LBL applied math seminar**, Organizer
2025	**Open problems and recent directions in matrix computations seminar**, Organizer
2022–2025	**Caltech CMS Keller colloqium committee**, Member
2024	**SIAM Mathematics of Data Science Conference, *Randomized matrix computations for large-scale scientific and machine learning problems* minisymposium**, Co-organizer
2023	**International Conference on Industrial and Applied Mathematics Conference, *Randomized numerical linear algebra* minisymposium**, Co-organizer

Service and Outreach

2021–2023	**Math "boot camp" for incoming PhD students**, Lecturer and co-organizer
2023	**AISTATS**, Top reviewer
2023	**SoME2 Math Exposition Contest**, Top 100 entry

Workshops

Fall 2025	**Simons workshop: Complexity and Linear Algebra**, invited participant
Mar. 2023	**BIRS workshop: Perspectives on Matrix Computations: Theoretical Computer Science Meets Numerical Analysis (23w5108)**, invited participant

Peer Review

SIAM Journal on Numerical Analysis (SINUM)
SIAM Journal on Scientific Computing (SISC)
SIAM Journal on Matrix Analysis and Applications (SIMAX)
SIAM Journal on Mathematics of Data Science (SIMODS)
Computational Science and Engineering
BIT Numerical Mathematics
Theory of Quantum Computing (TQC) Conference
Quantum Information Processing (QIP) Conference
Quantum Simulation (QSim) Conference
International Journal of Theoretical Physics
Artificial Intelligence and Statistics (AISTATS) Conference
Neural Information Processing Systems (NeurIPS) Conference
International Conference on Learning Representations (ICLR) Conference
ACM–SIAM Symposium on Discrete Algorithms (SODA) Conference
European Symposium on Algorithms (ESA) Conference

Professional Memberships

Society for Industrial and Applied Mathematics, 2018–present
Association for Computing Machinery, 2023–present